

POWER GENERATION SOLUTIONS · THE PLANT MODEL

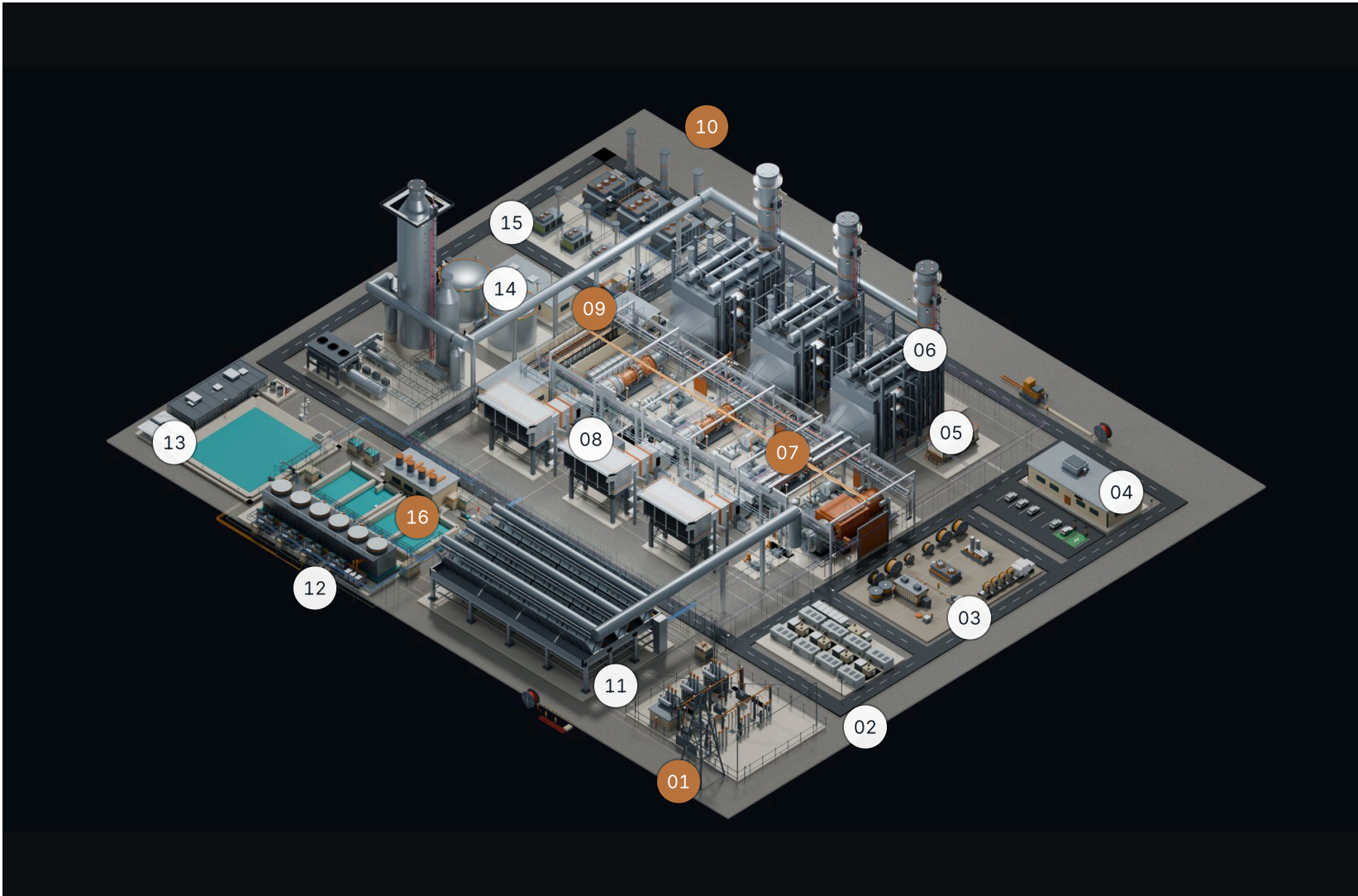
Accelerating Time to Power.

Cable, application engineering and project execution, from the generation asset to the grid interface.

[Start a cable package review](#)

One plant. Sixteen application zones.

Every zone is an equipment package, a cable scope and a buyer. The interactive e-brochure opens each one from the model.



Original engineering model. Illustrative; not a validated engineering design or a digital twin. Cable design and quantities follow the approved project scope.

01	Switchyard / GSU / grid tie
02	Battery energy storage
03	Reel staging / prefab
04	Admin / control room
05	E-house
06	HRSG trains & stacks
07	Turbine hall
08	GT air inlets
09	MCC / VFD / UPS
10	Modular power yard
11	Air-cooled condenser
12	Chillers / cooling towers
13	Water / wastewater
14	Carbon capture
15	Gas metering
16	Shared cable corridor

What has to be settled before the cable is released.

Eight decisions, drawn from the zone framework. Zone numbers follow the model.

ZONES 06 · 07 · 08

Heat, signals and OEM leads

Confirm hot-route exposure, oil/vibration, shield treatment and entry space. Keep CEMS heated umbilicals and heat-trace elements in their vendor packages; define separate plant feeders.

ZONES 11 · 12 · 13

Wet service has different duties

Distinguish washdown from continuous immersion; confirm chemicals, UV, starting voltage drop, drive lead length and motor termination. Specify true festoon/reeling duty separately.

ZONES 14 · 15

Account for the complete process

Add CCS power loads when process data is available. Obtain area classification, gas group, glands/seals and corrosion-control design. Hydrogen compatibility requires a defined system basis.

ZONES 04 · 05 · 09

Make power and controls fit

Align cable OD, bend space, conductor class, lugs and terminal limits. Define CT/VT circuits, shield bonding and diverse protection paths. Use listed circuit-integrity systems only for compatible fire-alarm/communications duty.

ZONES 02

Four systems within one fence

Fix DC polarity/fault duty, connector and crimp compatibility, collector voltage and the factory-to-field boundary. Container count does not establish cable footage; do not add internal battery harnesses by default.

ZONES 10

Match the operating mission

Distinguish standby, emergency and true black-start duty. Define load steps, essential-bus/ATS interfaces and independent routes. Separate permanent feeders from temporary load-bank, welding and maintenance leads.

ZONES 01

One agreed grid handoff

Confirm ownership, voltage, terminals, sheath bonding, ground-potential interfaces and diverse protection routes. Select OPGW or ADSS only when the overhead communications architecture requires it.

ZONES 03 · 16

Build the route into the package

Reconcile source-to-destination lengths, thermal conditions, fill, bends, pull direction, tension and sidewall pressure. Confirm end seals, access and firestop responsibility. Shared tray or duct length is not additional cable footage.

Seven families. One electrical scope.

Published constructions with specification numbers. Stock numbers do not indicate inventory; the technical appendix carries every reference and its manufacturer document.

6 CORE · 6 EXPANDED

Medium - voltage power

5/8 kV, 15 kV and 35 kV MV-105 constructions for plant feeders, source and tie circuits, collectors and transformer feeds. Select the construction from the actual system voltage, insulation level and routing design.

55052299 55050899 89008499 67747299
66131915 89066999

3 CORE · 4 EXPANDED

Low - voltage distribution and motor feeders

600 V and 600/1000 V EPR-insulated singles and multiconductor tray-rated feeders for secondaries, motors, panels and equipment.

58950101 / 58950104 58949999 59199899

3 CORE · 1 EXPANDED

VFD cable

Shielded drive-output constructions, flexible or armored, matched to the drive system. Qualify the drive before selecting the cable.

45834799 64346799 89052099

10 CORE · 7 EXPANDED

Control and instrumentation

CT/VT and breaker-control multiconductors, shielded pairs and triads for analog, status, RTD and thermocouple circuits, and armored instrumentation for classified areas. Type TC and TC-ER are distinct ratings.

59563399 59563699 59547599 59193199
577883 (base code) 59212899 56295201
89057289 / 89057399 59945899 59548199

2 CORE · 3 EXPANDED

Flexible, portable and approved DC applications

DLO flexible power for emergency and battery circuits, and RenewaFLEX for approved internal BESS wiring. Fixed wiring and portable cable are different applications; BESS DC is not solar PV.

167022R108 / 167024R108 13892403

1 CORE · 0 EXPANDED

Grounding and bonding

Bare copper and insulated equipment grounds for the ground grid, equipment bonds and shield grounds. Final sizes follow the grounding study.

10704542 / 16266909

4 CORE · 3 EXPANDED

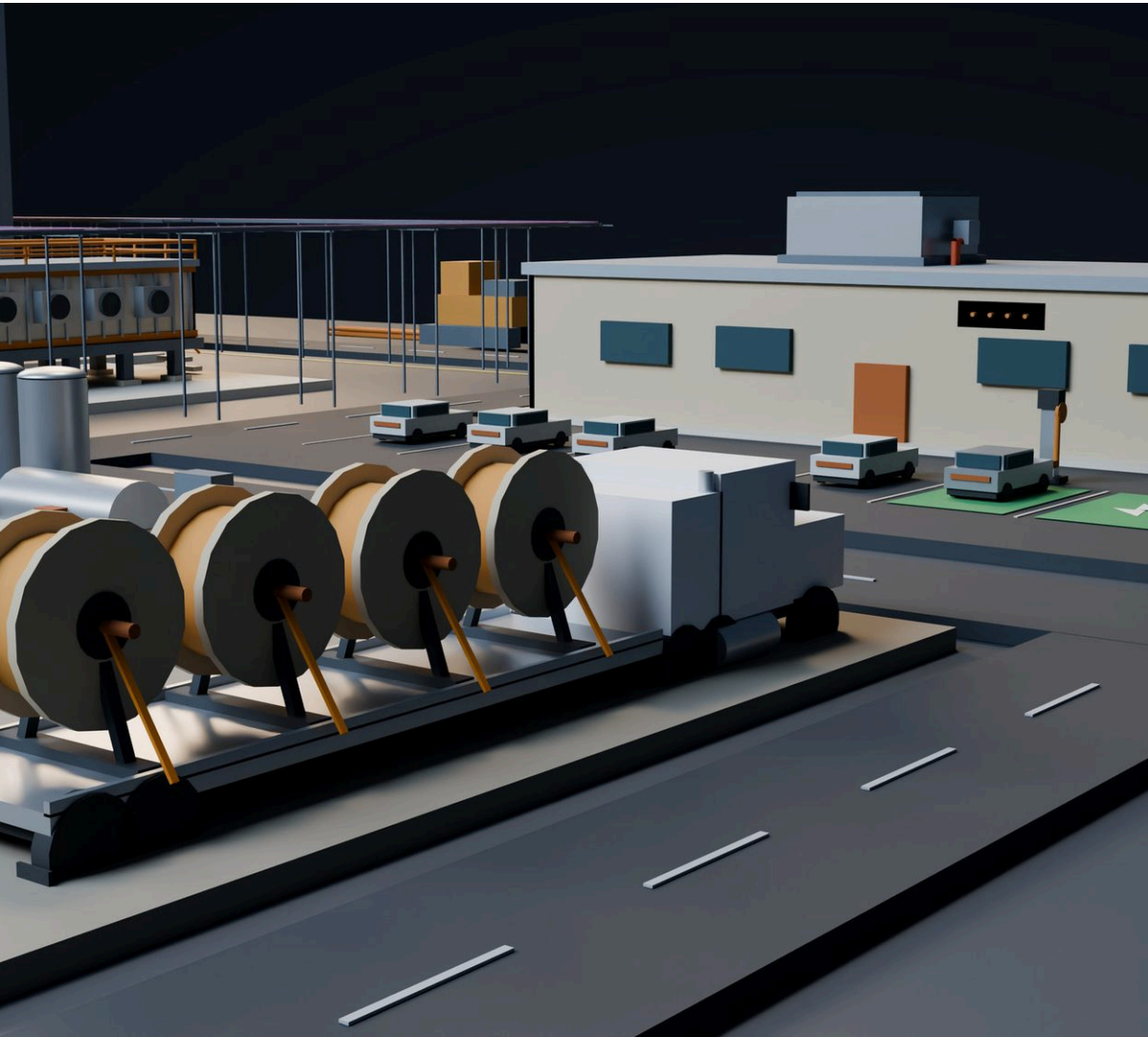
Communications, building systems and specialty

Fiber and industrial networks, building-system and branch wiring, switchboard wire, terminations and accessories, and OEM or special-environment leads that follow equipment specifications.

188040509 SIMpull Barrel THHN/THWN-2 Genesis
VFD spec sheets 85451 / 85452 / 85456

Make the schedule work in the field.

Package the delivery around the installation sequence: reel lengths, circuit identification, staged issue.



Reel staging and prefab, zone 03. Staging adds no circuit footage; it changes when and how the footage is issued.

PLAN &
PREPARE

Construction Planning Services

Takeoffs, BOMs, field drawings, coordination and pull calculations. Qualify circuit marking, pull heads and bundling against the installation sequence.

[Construction Planning Services](#) [Service overview · PDF](#)

SPECIFY

CableTechSupport

Application review for construction and application questions before the specification freezes.

[CableTechSupport](#)

PROCURE

SPEED Services

Qualified custom requests and schedule needs.

[SPEED Services](#)

DELIVER

Project Services

Coordination, marking, kitting, labels and bundles packaged around the installation sequence.

[Project Services](#)

DELIVER

Reel tracking

Qualify reel tracking, pull lengths and staged issue with the project team.

[Southwire Reel Tracking System](#)

INSTALL

Contractor Solutions & SIMpull

Bring the field team into route layouts, pull calculations and equipment setup. Align handling, pulling and crew preparation before cable arrives.

[Contractor Solutions](#) [SIMpull Cable-in-Conduit](#)

ASSESS &
SUSTAIN

Field Assessment & rejuvenation

Evaluate existing underground systems with Field Assessment Services. For eligible aging cable, assess rejuvenation alongside repair or replacement and the planned outage

[powergen@southwire.com](#)

[Field Assessment Services](#) [Cable rejuvenation](#)

Build the package around the application.

The combined-cycle model is a starting point. Adapt scope to the generation technology, site and operating duty.



CONVENTIONAL GENERATION	Combined cycle, simple cycle & peakers Plant MV distribution, turbine auxiliaries, HRSG instruments, cooling motors and switchyard protection. A simple-cycle plant has a different balance of plant.
MODULAR / BEHIND THE METER	RICE, fuel cells & onsite power Separate internal OEM wiring from yard feeders, starting/DC circuits, controls and grounding. For data centers, microgrids and industrial CHP, confirm the generation-to-campus handoff.
GENERATION + STORAGE	BESS & hybrid sites Distinguish internal battery wiring, field DC interconnects and AC collector feeders. Confirm voltage, polarity, terminals, fire requirements and the integrator's approved cable design.
EXISTING ASSETS / EXPANSION	Brownfield upgrades & new process loads Assess aging feeders and outage windows. Add a separate motor and instrument schedule for carbon capture, water treatment or cooling expansion; verify spare route and switchgear capacity.

ZONE 01 · PLANT TO GRID

Qualify the connection before selecting the cable.

Underground transmission

Southwire's published 69–500 kV scope includes cable systems, accessories, engineering, installation and testing. Qualify voltage, route, bonding, terminations and the utility interface.

[Products & services](#) · [Transmission line card](#) · [PDF](#)

Overhead gen-tie & grid upgrades

Compare conventional conductors with C7 composite-core options when thermal capacity, span, sag or reconductoring drives the design. Final selection belongs to the line engineer and utility.

[Conductor comparison](#) · [PDF](#)

32 miles

High-voltage cable, manufactured and installed. [Case study](#) · [PDF](#)

1969
vintage

A 69 kV underground circuit kept in service. [Case study](#) · [PDF](#)

Agree the generator step-up transformer (GSU), point of interconnection (POI), asset owner and route. Underground and overhead connections require different engineering packages.

Results with a reference.

Historical Southwire projects with their published sources. The PowerGen use is proposed, not an outcome from the illustrated plant.

CUSTOM MV CABLE / CANADIAN METALS & MINING

15 weeks

From RFQ to shipment.

Result. Southwire reports shipment in 15 weeks against a 20-week reference, with a 33% reduction in minimum order quantity.

PowerGen use, proposed. Bring cable engineering in early for unusual e-house, ACC or process-feeder constructions and quantities.

[Case study · PDF](#)

FEEDER LIFE EXTENSION / AES OHIO

42,255 ft

Rejuvenated at Dayton Airport.

Result. Southwire reports \$530,161 saved against estimated replacement cost, with the feeder project completed in two months.

PowerGen use, proposed. Assess rejuvenation for suitable aging MV feeders during brownfield life-extension planning.

[Case study · PDF](#)

FIELD INSTALLATION / BJC WEST COUNTY HOSPITAL

16 pulls

A coordinated installation setup.

Result. Southwire documents 16 pulls of 220–230 feet using 4 × 750 kcmil aluminum conductors with ground.

PowerGen use, proposed. Use pull calculations, reel setup and coordinated equipment planning for auxiliary feeders. Confirm the actual route and conductor construction.

[Case study · PDF](#)

More: [Onshore HV connection / Vineyard Wind 1 \(32 miles\)](#) · [URD reliability program / NV Energy \(\\$800 million\)](#) · [69 kV transmission / Atlantic Municipal Utilities \(1969 vintage\)](#).

Connect your next power project.

Start with the one - line diagram, the cable schedule, the equipment interfaces and the target energization date.

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Power Generation Solutions

Start a cable package review

01	Electrical basis One - line, system voltage, load and fault data, circuit schedule, grounding and redundancy requirements.
02	Route & environment Tray, duct or direct burial; lengths and bends; ambient heat, chemical exposure, wet locations and hazardous - area classification.
03	Interfaces & responsibility OEM terminal boundaries, GSU/POI limits, specification owner, package buyer, installer and testing responsibility.
04	Release & documentation Design - freeze and energization dates, reel and delivery plan, approved substitutions, test records and project - specific origin or environmental documentation.

Based on the PowerGen email brochure rev 5 (28 pages, September 2026) and the September 2026 strategy update, with linked Southwire references. The source states its manufacturer references were checked September 2026. Project engineering governs; final engineering and commercial terms are project - specific.